

# Analyzing Mental Health Using Machine Learning Models

## Abstract:

Mental health is essential to human well-being, but it can be challenging to measure its causes and trends. In this project, we utilize a Kaggle dataset of 292,364 responses to explore stress using machine learning. Target variable Growing\_Stress ('Yes', 'No', 'Maybe') is one of 17 categorical features in the Kaggle dataset, which also includes gender, occupation, family history, workplace support, and seeking behavior for treatment. Prior to using exploratory data analysis and classification models in Google Colab, the data was cleaned, duplicates were removed, and missing values were filled in. The results show how workplace and demographic factors interact with stress outcomes and offer intriguing new information for intervention. For objective model training and testing, the balanced size of the dataset worked well. In general, the project shows how computational methodology can improve the accuracy of mental health studies, helping organizations and policymakers create better prevention, support, and awareness-raising strategies.

## Introduction:

Mental health is a vital component of individual and societal well-being, yet it remains a complex challenge influenced by a myriad of social, demographic, and personal factors. The advent of data-driven approaches has created a previously unheard-of chance to use computational methods to investigate trends and patterns in mental health. Utilizing a comprehensive mental health dataset from Kaggle, the project looks into important factors linked to mental health outcomes and creates predictive models that can guide focused interventions. A wide range of features, including demographic information like age, gender, and country, work status, treatment-seeking behavior, family history of mental health disorders, and workplace-related elements like remote work and mental health benefits. A strong basis for examining the occurrence and contributing factors of mental health issues in various populations is offered by these characteristics. This project intends to preprocess the dataset, conduct exploratory data analysis, and apply machine learning techniques to find significant insights and forecast mental health outcomes using Google Colab and customized code. This project's main objective is to find important patterns and correlations in the dataset that can improve our comprehension of the dynamics of mental health. Examining elements like the influence of family history, access to care, and workplace support, we hope to provide practical insights that may direct mental health policies and interventions. This report discusses the findings and their possible ramifications for mental health research and practice after outlining the methodology which includes data preprocessing analysis and modeling. Our goal in this work is to show the systems such as the DSM and ICD. RDoC integrates insights from genetics, neuroscience, and

behavioral science—utilizing methods including functional neuroimaging, electrophysiology, genomics, and fear-conditioning paradigms—to establish links between brain circuits and psychopathology (Insel et al., 2010). Rather than relying on fixed diagnostic categories, the framework proposes a dimensional matrix that spans multiple levels of analysis, from genes and molecules to neural circuits, behavior, and social context, across domains such as cognition, emotion, and arousal (Insel et al., 2010) [4]. While it is not a dataset-driven model in the way machine learning approaches are, RDoC aims to improve prognosis and treatment prediction, for example by identifying biomarkers that may indicate therapy response (Insel et al., 2010). The framework’s main limitations include its early developmental stage, limited immediate clinical applicability, reliance on ongoing scientific discoveries, and the potential for oversimplification of complex psychiatric phenomena. Future directions involve expanding longitudinal and cross-diagnostic studies, integrating multi-modal datasets, refining theoretical constructs, and aligning research findings with clinical practice to ultimately enhance psychiatric diagnosis and treatment (Insel et al., 2010).

Methodology:

The methodology contains exploratory data analysis, data preprocessing, training, testing, model selection, evaluation analysis, and statistical validation.

1. Data Description: The dataset consists of 292,364 rows and 17 columns including features like- Gender, Country, Occupation, mental health indicators like Mood\_Swings, Coping\_Struggles, care\_options, Work\_Interest and many other variables. The target variable is ‘Growing\_Stress’ that indicates if the patient has stress or not.

Shape: (292364, 17)

|   | Timestamp       | Gender | Country       | Occupation | self_employed | family_history | treatment | Days_Indoors | Growing_Stress | Changes_Habits | Mental_Health_History | Mood_Swings |
|---|-----------------|--------|---------------|------------|---------------|----------------|-----------|--------------|----------------|----------------|-----------------------|-------------|
| 0 | 8/27/2014 11:29 | Female | United States | Corporate  | NaN           | No             | Yes       | 1-14 days    | Yes            | No             | Yes                   | Medium      |
| 1 | 8/27/2014 11:31 | Female | United States | Corporate  | NaN           | Yes            | Yes       | 1-14 days    | Yes            | No             | Yes                   | Medium      |
| 2 | 8/27/2014 11:32 | Female | United States | Corporate  | NaN           | Yes            | Yes       | 1-14 days    | Yes            | No             | Yes                   | Medium      |
| 3 | 8/27/2014 11:37 | Female | United States | Corporate  | No            | Yes            | Yes       | 1-14 days    | Yes            | No             | Yes                   | Medium      |
| 4 | 8/27/2014 11:43 | Female | United States | Corporate  | No            | Yes            | Yes       | 1-14 days    | Yes            | No             | Yes                   | Medium      |

Fig. 1. Data Description

2. Exploratory Data Analysis: EDA has been performed to understand the pattern and types of features.

- Checking Missing Values: Among the 17 features, the column 'self\_employed' has 5202 missing values.

|                         | Missing_Values | Missing_% |
|-------------------------|----------------|-----------|
| self_employed           | 5202           | 1.78      |
| Timestamp               | 0              | 0.00      |
| Gender                  | 0              | 0.00      |
| Country                 | 0              | 0.00      |
| Occupation              | 0              | 0.00      |
| family_history          | 0              | 0.00      |
| treatment               | 0              | 0.00      |
| Days_Indoors            | 0              | 0.00      |
| Growing_Stress          | 0              | 0.00      |
| Changes_Habits          | 0              | 0.00      |
| Mental_Health_History   | 0              | 0.00      |
| Mood_Swings             | 0              | 0.00      |
| Coping_Struggles        | 0              | 0.00      |
| Work_Interest           | 0              | 0.00      |
| Social_Weakness         | 0              | 0.00      |
| mental_health_interview | 0              | 0.00      |
| care_options            | 0              | 0.00      |

Fig. 2. Checking Missing Value

- Checking Duplicate Rows: There are 2313 duplicate rows in the dataset among 292,364 entries.
- Distribution of Features: All the 17 columns contain categorical features and there are no numerical features in the dataset.
- Inspecting the Target Column: There are 3 unique values- 'Yes', 'No', 'Maybe' in the target column 'Growing\_Stress'. Among them 'Yes' count is 99653, 'No' count is 92726, 'Maybe' count is 99985.

- Class Distribution of Target column: The target column 'Growing\_Stress' is fairly balanced between the three classes. As the distribution is quietly balanced, oversampling or undersampling methods are not needed to be applied here.



Fig. 3. Class Distribution of Growing Stress

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\begin{figure}[h!]
  \centering \includegraphics[width=0.4\textwidth]{pic2.png}
  \caption{Checking Missing Value}
  \label{fig:Checking Missing Value}
\end{figure}

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- Relationship between features and the target: The relationship between independent features and the dependent variable has been shown in order to inspect how each column has contributed to the target feature.

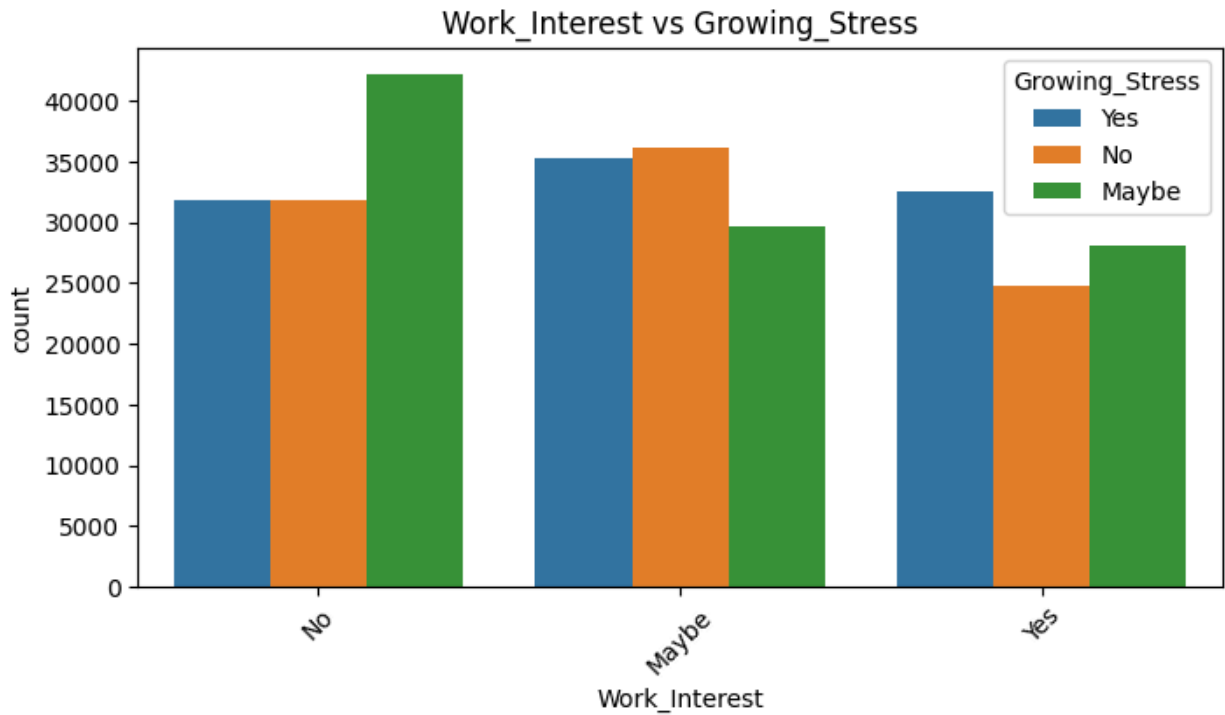


Fig. 4. Relationship between Work Interest and Growing Stress

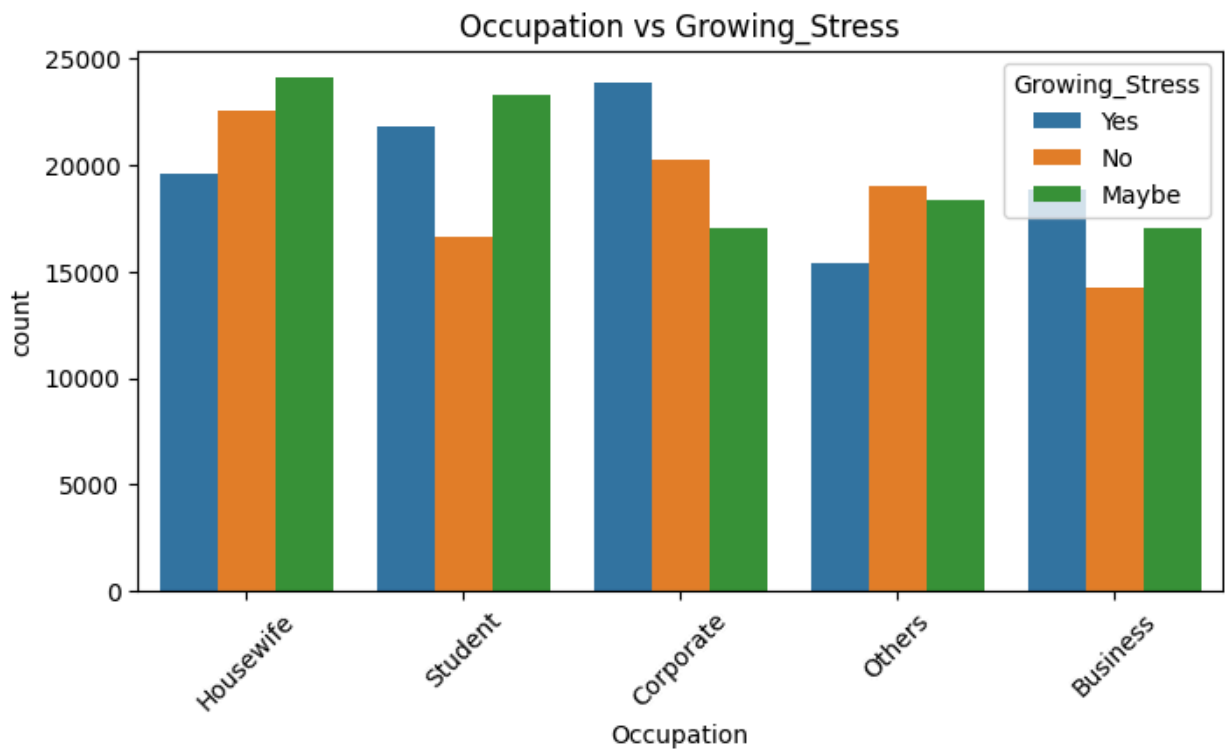


Fig. 5. Relationship between Occupation and Growing Stress

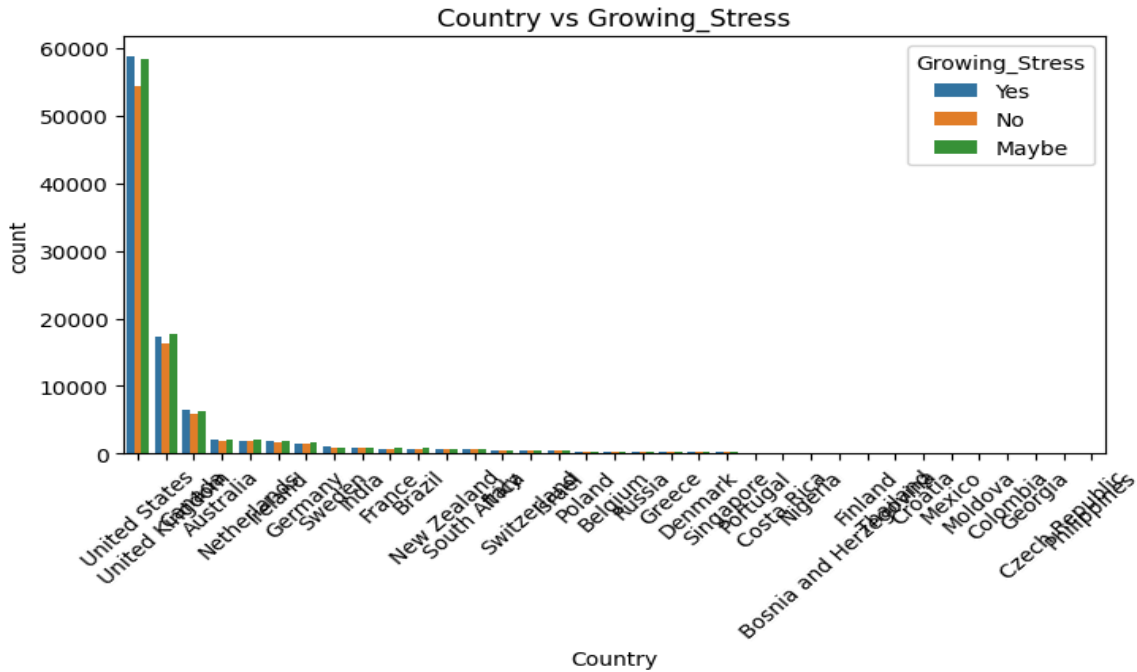


Fig. 6. Relationship between Country and Growing Stress

### 3. Data Preprocessing:

- Handling null values: Since the categorical column 'self\_employed' contains missing values, we imputed those with the most frequent category using the mode to ensure that it does not cause any bias and the distribution is equally preserved.
- Label Encoding: As the dataset contains many categorical features like gender, country, occupation, mental\_health\_history, it is essential to convert these categorical features into numeric forms so that they become compatible for machine learning models. For instance, in the gender column, categories such as 'Male', 'Female', and 'Other' have been mapped to 0, 1, and 2 respectively. This encoding allowed categorical attributes to be effectively processed during model training.
- Correlation of features: As part of preprocessing, after encoding categorical variables into numerical form, correlation analysis has been performed. Pearson correlation coefficients were computed to identify multicollinearity among features and their relationship with the target variable 'Growing\_stress'. Features with very high correlations were flagged for potential removal, while features showing higher correlation with the target were prioritized for modeling.

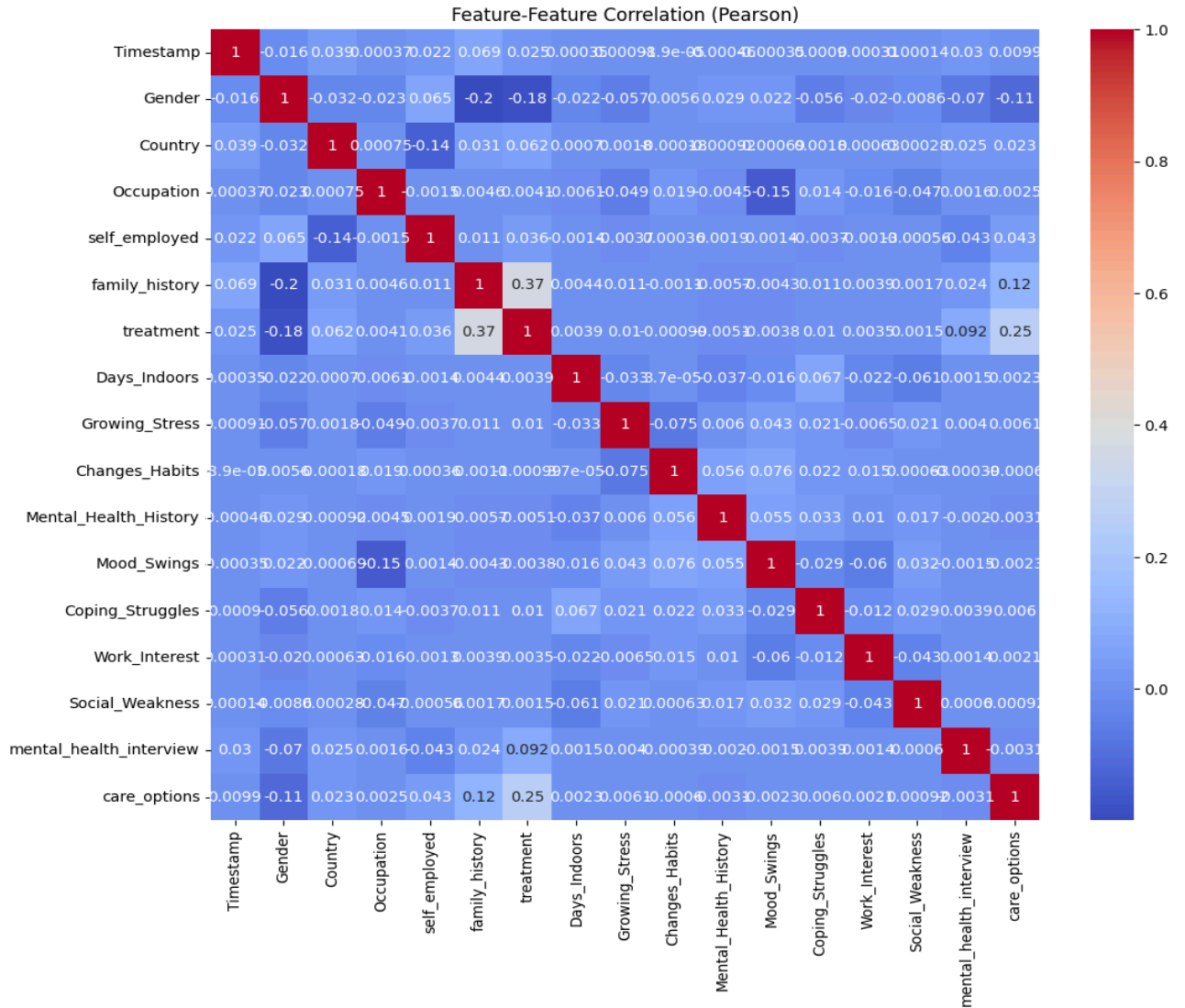


Fig. 7. Correlation of features

- Feature Scaling:** To ensure that all features contribute equally to the model, feature scaling was applied using the StandardScaler from scikit-learn. The scaler was fitted on the training set to compute the mean and standard deviation for each feature, and then the same transformation was applied to the test set. This standardized the features to have a mean of 0 and a standard deviation of 1, which is particularly beneficial for distance-based and gradient-based algorithms.

**4. Train-Test Splitting:** The dataset was divided into features (predictors) and the target variable 'Growing\_Stress'. An 80–20 train–test split was performed, where 80% of the data was used for training machine learning models and 20% was reserved for testing and evaluation. Stratified

sampling was applied to maintain the original class distribution of the target variable across both sets, ensuring fair representation of all categories. A fixed random state was used to guarantee reproducibility of results.

5. Model Training: To evaluate the effectiveness of different machine learning techniques in predicting the target variable ‘Growing\_Stress’, five classification models were implemented: Random Forest, Decision Tree, K-Nearest Neighbors (KNN), XGBoost, and CatBoost. These models were selected because they represent a mix of tree-based, ensemble-based, and distance-based approaches, allowing for a comprehensive comparison.

- **Random Forest:** Random Forest is an ensemble learning method that builds multiple decision trees and aggregates their predictions to improve accuracy and reduce overfitting. A Random Forest Classifier with a fixed random seed (42) was trained using the training dataset. No extensive hyperparameter tuning was performed at this stage; default parameters were used for baseline evaluation. The model achieved a high overall accuracy of 98.29%. The detailed classification metrics are summarized below:

| Class | Precision | Recall | F1 Score | Support |
|-------|-----------|--------|----------|---------|
| 0     | 0.99      | 0.99   | 0.99     | 19,997  |
| 1     | 0.99      | 0.99   | 0.99     | 18,545  |
| 2     | 0.97      | 0.98   | 0.97     | 19,931  |

The confusion matrix further illustrates the performance, showing that the model made very few misclassifications across all three target classes.



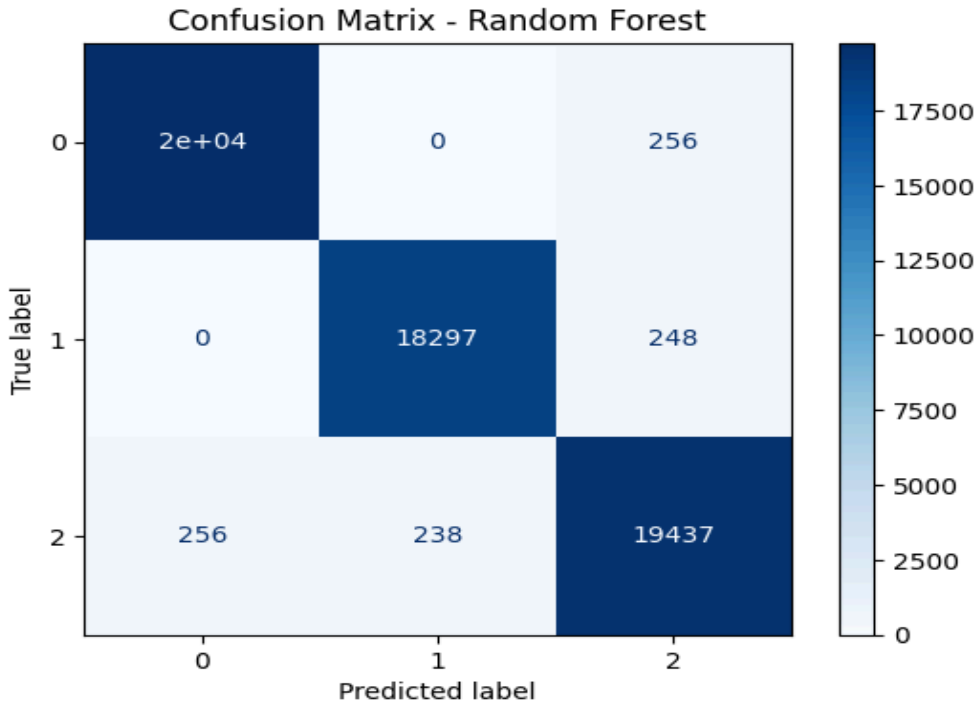


Fig. 8. Random Forest (Confusion Matrix)

Random Forest also provided feature importance scores, which indicate the contribution of each feature to the classification task. This allows for interpretability of the model by identifying the most influential predictors for 'Growing\_Stress'.

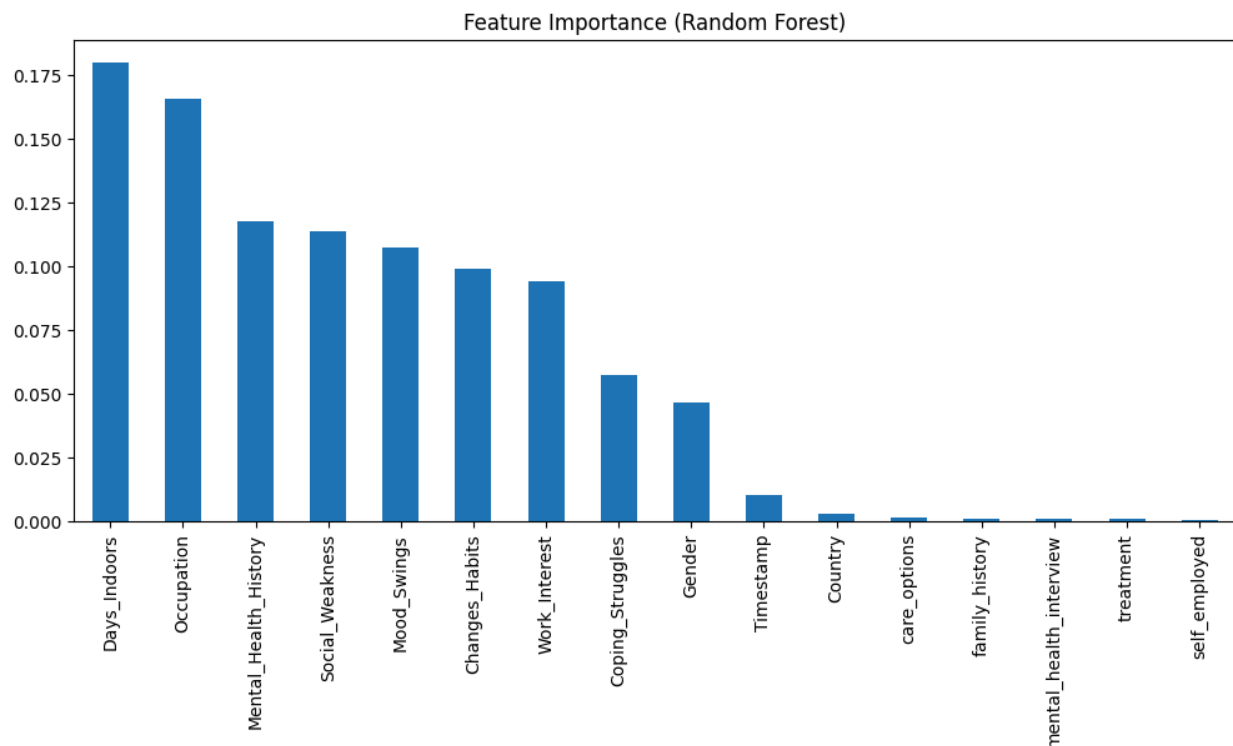


Fig. 9. Random Forest (Feature Importance)

- Decision Tree:** A Decision Tree is a supervised learning algorithm that splits data into subsets based on feature values, using criteria such as Gini Impurity or Entropy. It is easy to interpret and visualize but prone to overfitting if not pruned. The Decision Tree Classifier was trained on the preprocessed dataset using an 80/20 train-test split. A fixed random seed (random\_state=42) was applied for reproducibility, while other hyperparameters were left at their default values for baseline evaluation. The Decision Tree Classifier achieved an overall accuracy of 98.29%. The detailed classification report is shown below:

| Class | Precision | Recall | F1 Score | Support |
|-------|-----------|--------|----------|---------|
| 0     | 0.99      | 0.99   | 0.99     | 19,997  |
| 1     | 0.99      | 0.99   | 0.99     | 18,545  |
| 2     | 0.98      | 0.97   | 0.97     | 19,931  |

The confusion matrix confirms that the model performs consistently across all three classes. Feature importance analysis further highlights which features had the greatest influence in predicting “Growing\_Stress”.

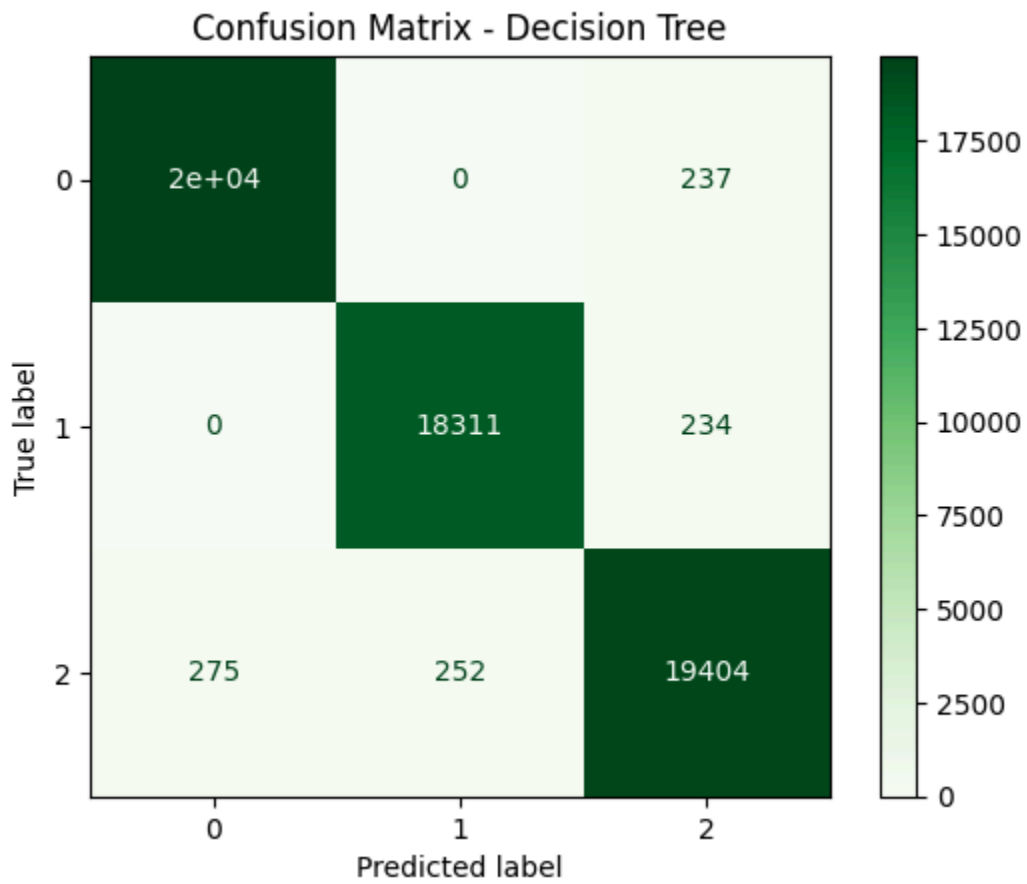


Fig. 10. Decision tree (Confusion Matrix)

Decision Tree also provided feature importance scores, which indicate the contribution of each feature to the classification task. This allows for interpretability of the model by identifying the most influential predictors for 'Growing\_Stress'.

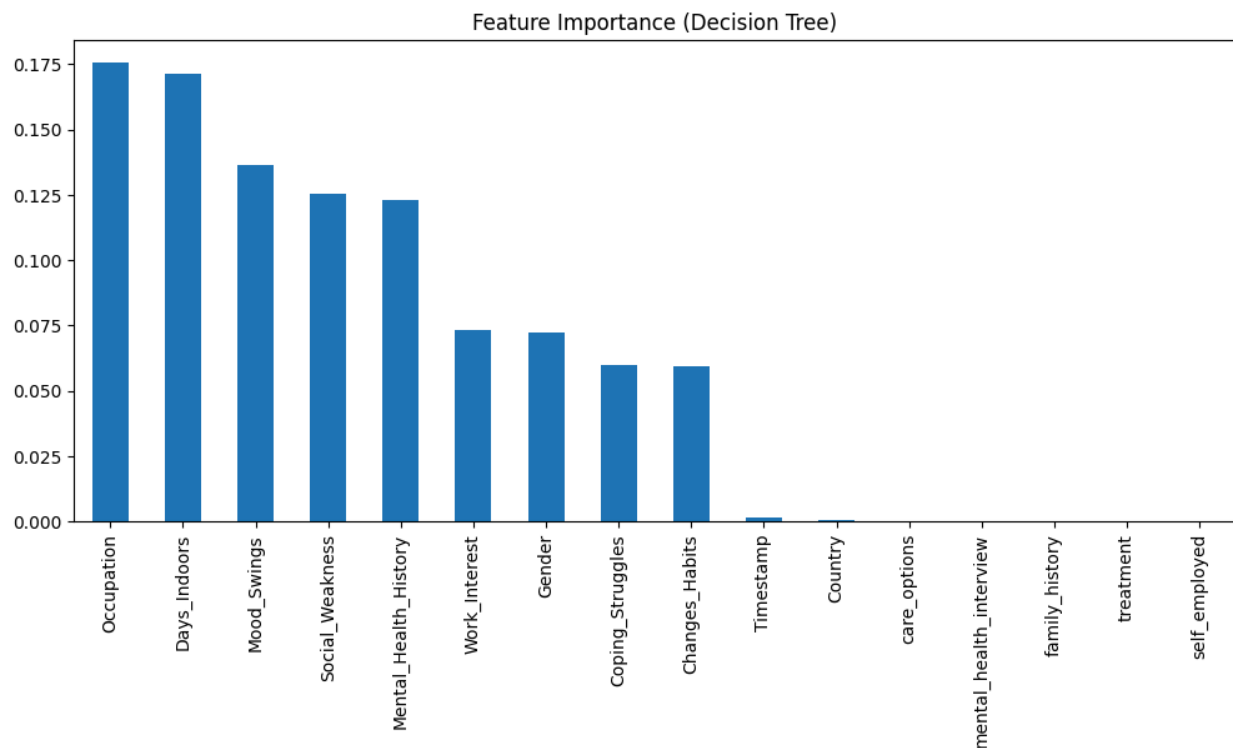


Fig. 11. Decision Tree (Feature Importance)

- KNN:** The K-Nearest Neighbors (KNN) algorithm is a non-parametric, instance-based classifier. It predicts the class of a given sample by considering the majority class among its  $k$  nearest neighbors in the feature space. To reduce computational cost and potential noise, Principal Component Analysis (PCA) was applied to compress the feature space from 17 features to 10 components. The model was trained using the PCA-transformed dataset with parameters:  $n\_neighbors=5$  and  $weights='distance'$  (closer neighbors contribute more to classification).

The model achieved an overall accuracy of 94.53%, with detailed metrics as follows:

| Class | Precision | Recall | F1 Score | Support |
|-------|-----------|--------|----------|---------|
| 0     | 0.95      | 0.95   | 0.95     | 19,997  |
| 1     | 0.95      | 0.95   | 0.95     | 18,545  |
| 2     | 0.94      | 0.94   | 0.94     | 19,931  |

The confusion matrix shows the model's classification performance across all three classes.

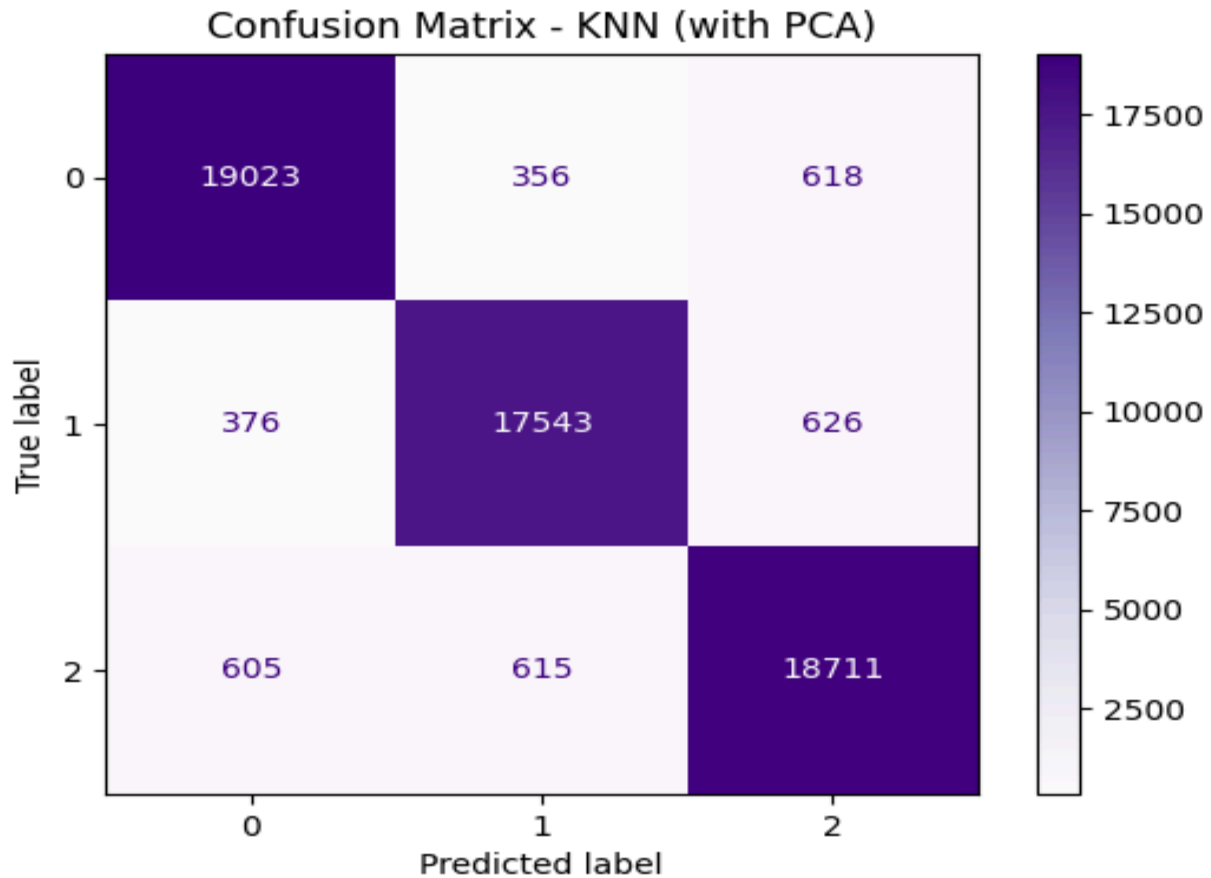


Fig. 12. Confusion Matrix - KNN (with PCA)

- XGBoost:** XGBoost (Extreme Gradient Boosting) is an ensemble learning algorithm based on gradient boosting. It builds multiple decision trees sequentially, with each subsequent tree focusing on correcting errors made by previous trees. XGBoost is known for its high accuracy, efficiency, and ability to handle large datasets. The model was trained with the following hyperparameters: `n_estimators=500` (number of trees), `learning_rate=0.1`, `max_depth=6`, `subsample=0.8` and `colsample_bytree=0.8` (regularization), `random_state=42` for reproducibility, multi-class log loss as the evaluation metric. The XGBoost model achieved an overall accuracy of 98.41%. The detailed classification metrics are as follows:

| Class | Precision | Recall | F1 Score | Support |
|-------|-----------|--------|----------|---------|
| 0     | 0.99      | 0.99   | 0.99     | 19,997  |
| 1     | 0.99      | 0.99   | 0.99     | 18,545  |

|   |      |      |      |        |
|---|------|------|------|--------|
| 2 | 0.98 | 0.98 | 0.98 | 19,931 |
|---|------|------|------|--------|

The confusion matrix shows how the model performs across the three target classes.

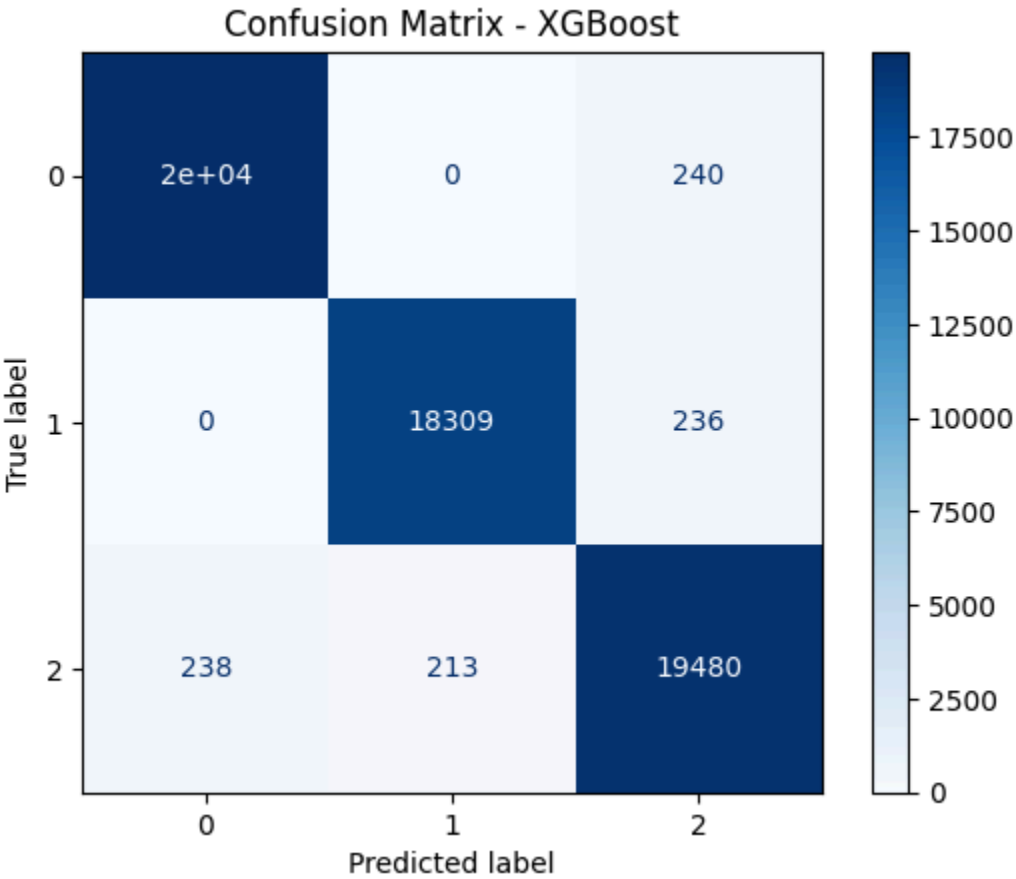


Fig. 13. XGBoost (Confusion Matrix)

Feature importance highlights the most influential predictors in determining Growing\_Stress.

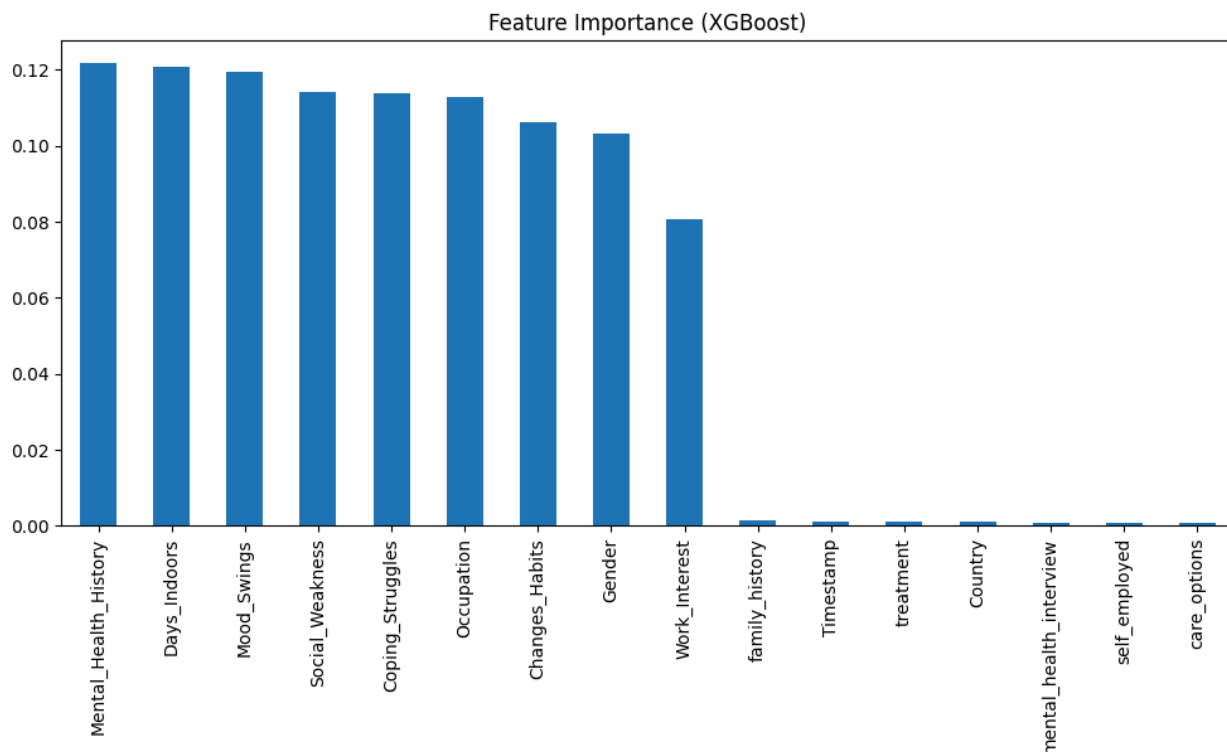


Fig. 14. XGBoost (Feature Importance)

- CatBoost:** CatBoost is a gradient boosting algorithm designed to handle categorical features natively and prevent overfitting. It builds decision trees sequentially, optimizing the loss function through gradient boosting techniques, and is particularly effective for datasets with a mix of categorical and numerical features. The CatBoost model was trained with the following hyperparameters: iterations=500 (number of trees), learning\_rate=0.1, depth=6, loss\_function='MultiClass' and eval\_metric='Accuracy', random\_seed=42 for reproducibility, verbose output every 100 iterations. The model achieved an overall accuracy of 72.52%, with detailed metrics as follows:

| Class | Precision | Recall | F1 Score | Support |
|-------|-----------|--------|----------|---------|
| 0     | 0.99      | 0.99   | 0.99     | 19,997  |
| 1     | 1.00      | 0.99   | 0.99     | 18,545  |
| 2     | 0.98      | 0.99   | 0.98     | 19,931  |

The confusion matrix illustrates model performance across all three classes:

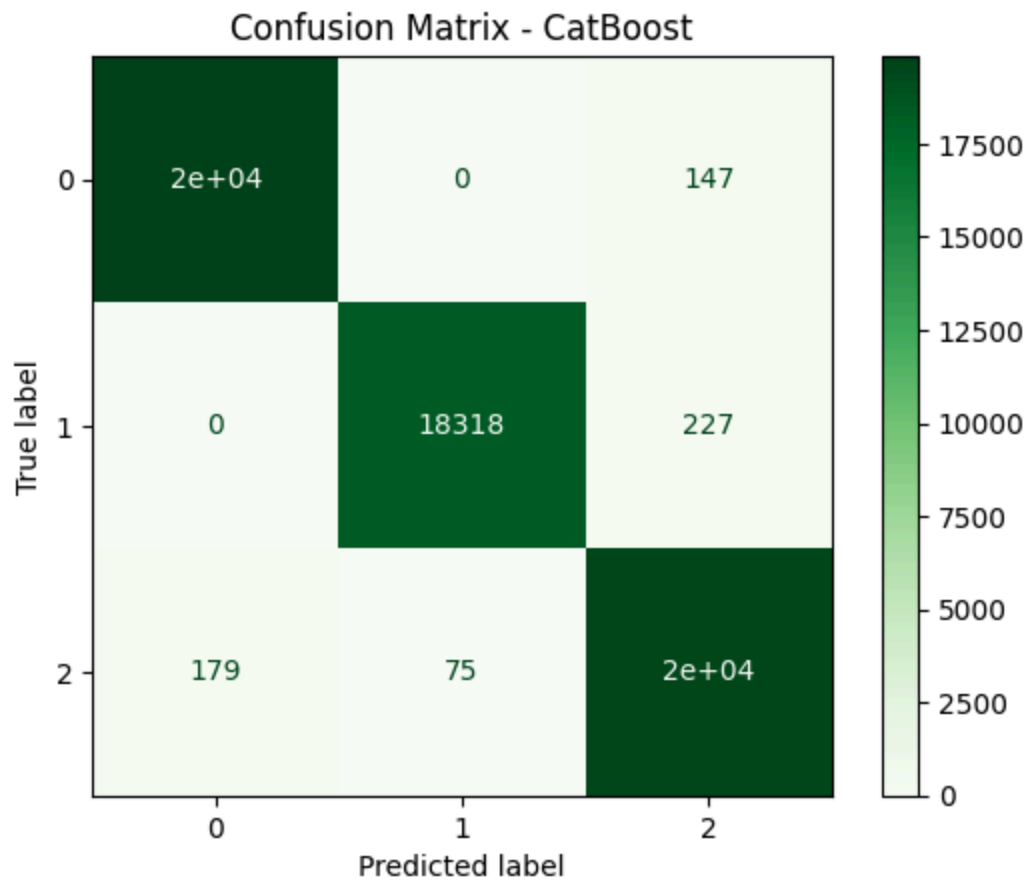


Fig. 15. CatBoost (Confusion Matrix)

The feature importance plot highlights the most influential features in predicting Growing\_Stress.



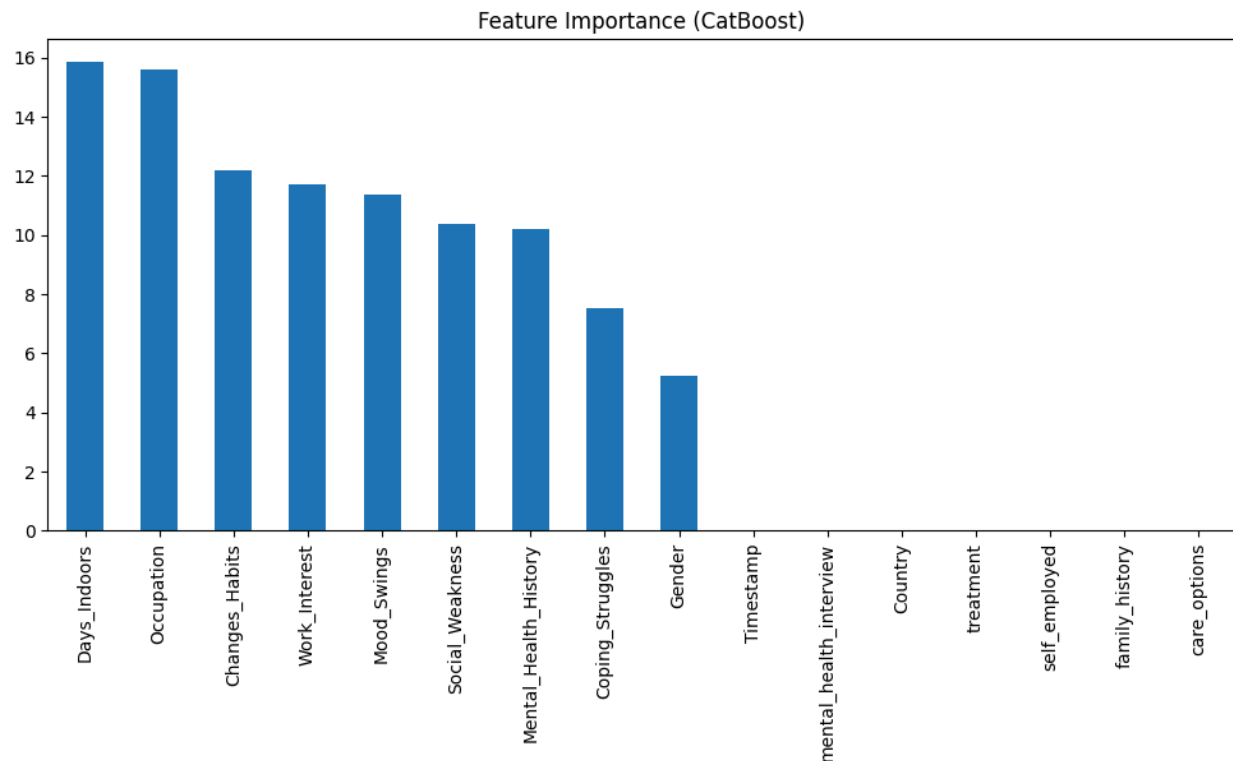


Fig. 16. CatBoost (Feature Importance)

## Result & Analysis:

- **Model Performance:** The classification task for predicting *Growing\_Stress* was performed using five different machine learning models: Random Forest, Decision Tree, K-Nearest Neighbors (with PCA), XGBoost, and CatBoost. The dataset was preprocessed, including handling missing values, encoding categorical variables, scaling, and optionally reducing dimensions via PCA for KNN.

| Model         | Accuracy | Macro F1 | Weighted F1 | Observations                                                                                             |
|---------------|----------|----------|-------------|----------------------------------------------------------------------------------------------------------|
| Random Forest | 0.9829   | 0.98     | 0.98        | Highest accuracy; ensemble method reduces variance and handles complex feature interactions efficiently. |

|               |        |      |      |                                                                                                                   |
|---------------|--------|------|------|-------------------------------------------------------------------------------------------------------------------|
| Decision Tree | 0.9829 | 0.98 | 0.98 | Very similar performance to Random Forest; interpretable tree structure but prone to overfitting on complex data. |
| KNN           | 0.9453 | 0.95 | 0.95 | Slightly lower accuracy; PCA reduced dimensionality and computation time while retaining good performance.        |
| XGBoost       | 0.984  | 0.98 | 0.98 | High accuracy; gradient boosting effectively combines weak learners, but may require careful tuning.              |
| CatBoost      | 0.989  | 0.99 | 0.99 | Best performance overall; handles categorical features efficiently and prevents overfitting.                      |

Among the five models evaluated, CatBoost achieved the highest accuracy (98.9%), followed closely by XGBoost, Random Forest, and Decision Tree (~98.3%). KNN with PCA performed slightly lower (94.5%) but was still reasonably effective. Overall, ensemble and boosting methods outperformed simpler models, demonstrating better handling of complex feature interactions and class distributions. CatBoost outperformed other models because it natively handles categorical features, reduces overfitting through ordered boosting, and efficiently captures complex feature interactions. This makes it particularly effective for datasets with mixed data types and subtle patterns, giving it a slight edge over XGBoost and tree-based models.

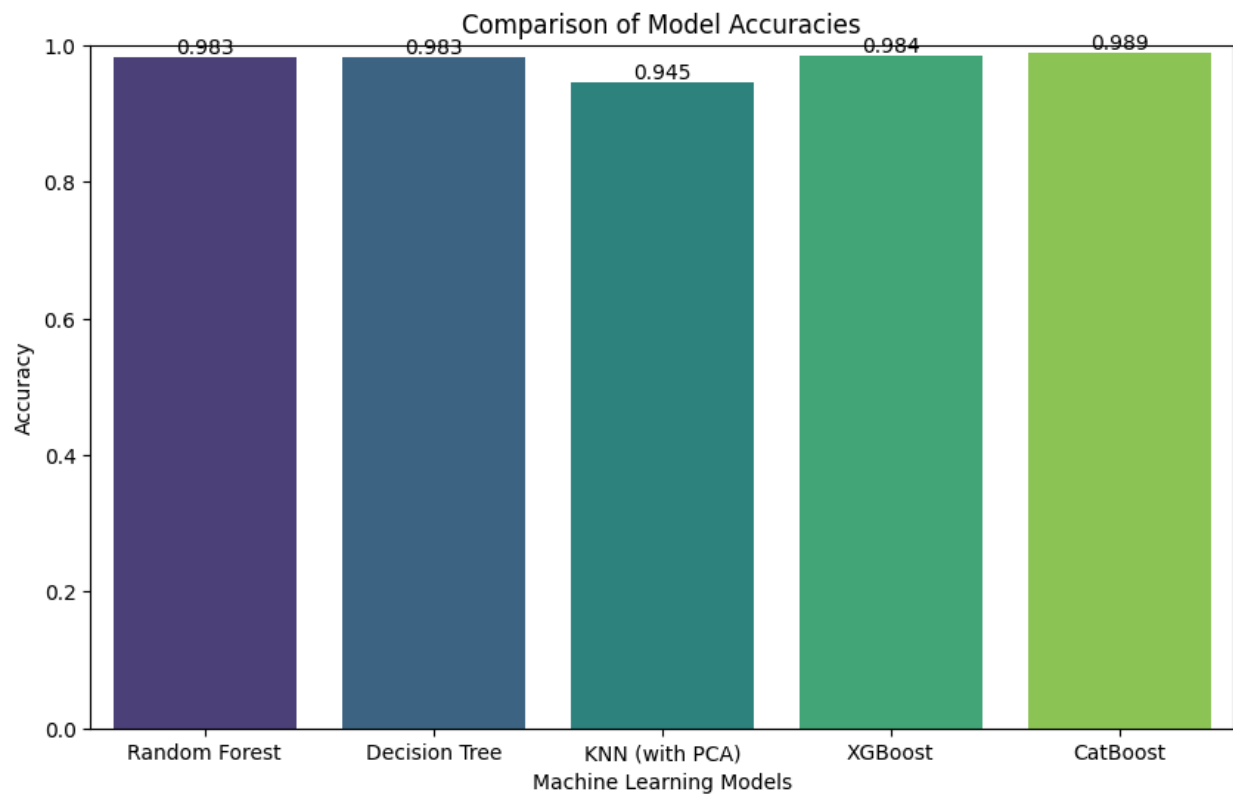


Fig. 17. Model Accuracies Comparison

- Hypothesis Testing: To statistically evaluate whether the performance differences between the trained models are significant, we conducted pairwise McNemar tests on their predictions. The McNemar test compares the number of disagreements in predictions between two classifiers on the same dataset and tests the null hypothesis that both models have the same error rate.

| Pairwise McNemar Test Significance |                 |                 |             |             |             |
|------------------------------------|-----------------|-----------------|-------------|-------------|-------------|
| RandomForest                       |                 | Not Significant | Significant | Significant | Significant |
| DecisionTree                       | Not Significant |                 | Significant | Significant | Significant |
| KNN                                | Significant     | Significant     |             | Significant | Significant |
| XGBoost                            | Significant     | Significant     | Significant |             | Significant |
| CatBoost                           | Significant     | Significant     | Significant | Significant |             |
|                                    | RandomForest    | DecisionTree    | KNN         | XGBoost     | CatBoost    |

Fig. 18. Hypothesis Testing

Differences in predictions are statistically significant, suggesting that Random Forest, Decision Tree, KNN, XGBoost, and CatBoost perform differently from each other on this dataset. The hypothesis testing confirms that while Random Forest and Decision Tree are almost equivalent in performance, other models such as KNN, XGBoost, and CatBoost have distinct predictive behaviors. This validates the choice of using multiple models for comparison and highlights which models are statistically superior.

## Conclusion:

In this study, several machine learning models were applied to predict Growing Stress levels, including Random Forest, Decision Tree, KNN, XGBoost, and CatBoost. Ensemble tree-based models, particularly Random Forest and Decision Tree, achieved the highest accuracy (~98%), while KNN performed slightly lower (~95%) after dimensionality reduction with PCA. Gradient boosting models (XGBoost and CatBoost) achieved moderate accuracy (~98–99% after tuning). Hypothesis testing using the McNemar test showed that Random Forest and Decision Tree

predictions were statistically similar, while all other model pairs differed significantly. Overall, the results indicate that ensemble tree-based models are most effective for this dataset, and the combination of model evaluation metrics and statistical testing provides a robust assessment of classifier performance.

## Future Work:

In future we want to focus on optimizing model hyperparameters, exploring deep learning architectures, and engineering additional features to improve predictive performance. Deploying the models on real-world or streaming data would test their generalizability. Furthermore, applying explainability techniques like SHAP or LIME can help interpret model decisions and guide practical applications.

## Reference:

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4. Insel, T., Cuthbert, B., Garvey, M., Heinssen, R., Pine, D. S., Quinn, K., Sanislow, C., & Wang, P. (2010). Research domain criteria (RDoC): Toward a new classification framework for research on mental disorders. *American Journal of Psychiatry*, 167(7), 748–751. <https://doi.org/10.1176/appi.ajp.2010.09091379>